

Orders

Previous Lecture

- *Describing binary relations*
- *Types of relations*
- *Equivalence relations*

Types of binary relations

● Reflexive

$(a,a) \in R$ for all $a \in A$.

● Symmetric

If $(a,b) \in R$ then $(b,a) \in R$.

● Transitive

If $(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$.

● Anti-symmetric

If $(a,b) \in R$ and $(b,a) \in R$ then $a=b$.

● Equivalence

If R is reflexive, symmetric, and transitive

Equivalence relations

● Recall that equivalence relations generalize the notion of *equality*

- For every a , a is equal to itself (*reflexive*)
- For every a and b , a and b are equal if b and a are equal (*symmetric*)
- For every a, b , and c , if a and b are equal and b and c are equal, a and c should be equal (*transitive*)

● How might we generalize the notion of an *ordering*?



If a comes before b , what can (or can't) be true?

If a comes before b and b comes before c , what do we know?

What about a ? Should it be “before” itself? “before or equal?”

Orders

- A relation R on a set A is called a **partial order** if it is **reflexive, transitive, anti-symmetric**.
- If R is instead **antireflexive**, then it is a **strict partial order**

Examples:

- $a \leq b$ on the set of real numbers is a partial order
 - $a < b$ on the set of real numbers is a **strict** partial order
 - is $a > b$ a (strict) partial order?
-
- A set S together with a partial order R is called a **poset** (partially ordered set)

Subsets

● **Claim:** $R = \{(a,b) \in P(A) \times P(A) \mid a \subseteq b\}$ is a partial order

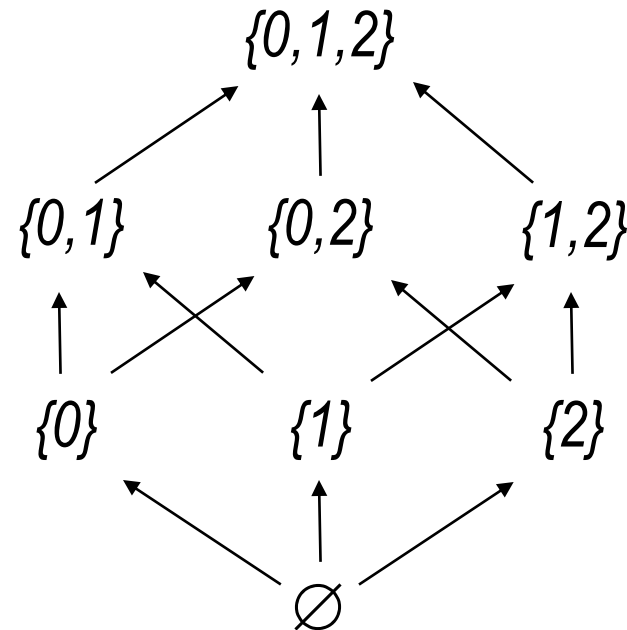
Reflexive?

Transitive?

Anti-symmetric?

Hasse diagrams

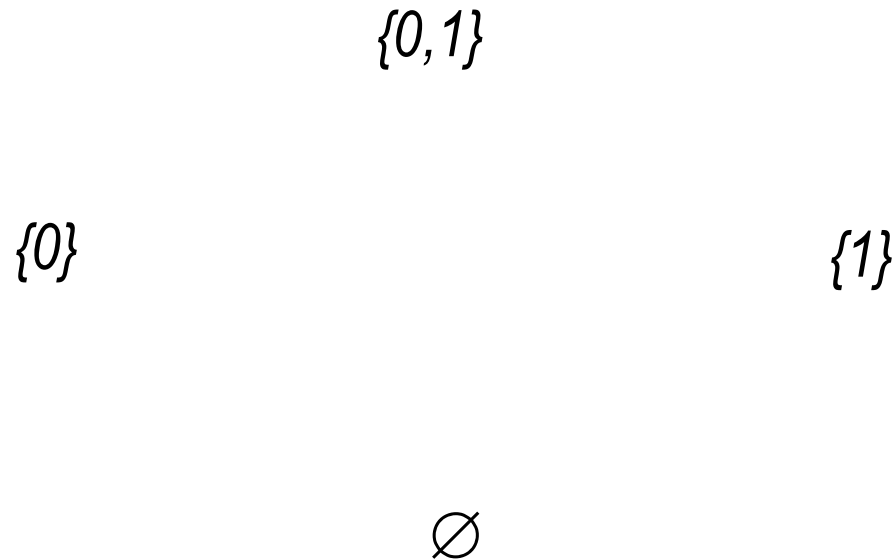
- *Standard way of drawing or visualizing a partial order is via a **Hasse diagram***



- *Idea is to **only show enough orderings** to reconstruct the whole order*
 - *Do not need to show reflexive edges*
 - *Do not need to show any edges implied by transitivity*
 - *Put elements higher in the order higher on the page, by anti-symmetry*

Constructing Hasse diagrams

- *Start with the whole order*
- *Remove any reflexive edges*
- *Remove any edges implied by transitivity*



Divisibility

- **Definition:** Given integers a and b , we say that a divides b (denoted $a \mid b$) if and only if $b = a \cdot k$ for some integer k

Does $3 \mid 6$?

Does $3 \mid 7$?

- **Claim:** $R = \{(a,b) \mid a \text{ divides } b\}$ is a partial order

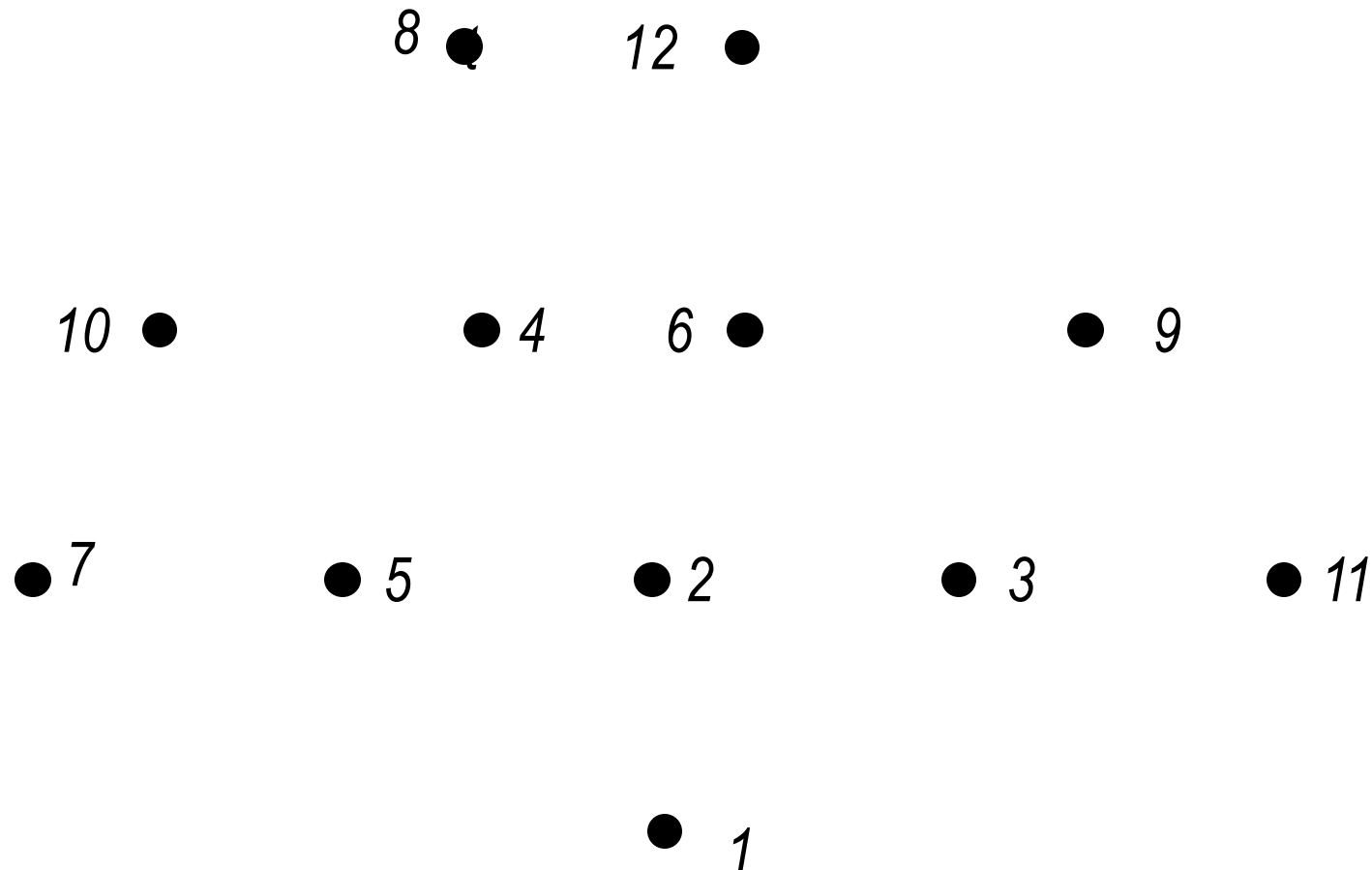
Reflexive?

Transitive?

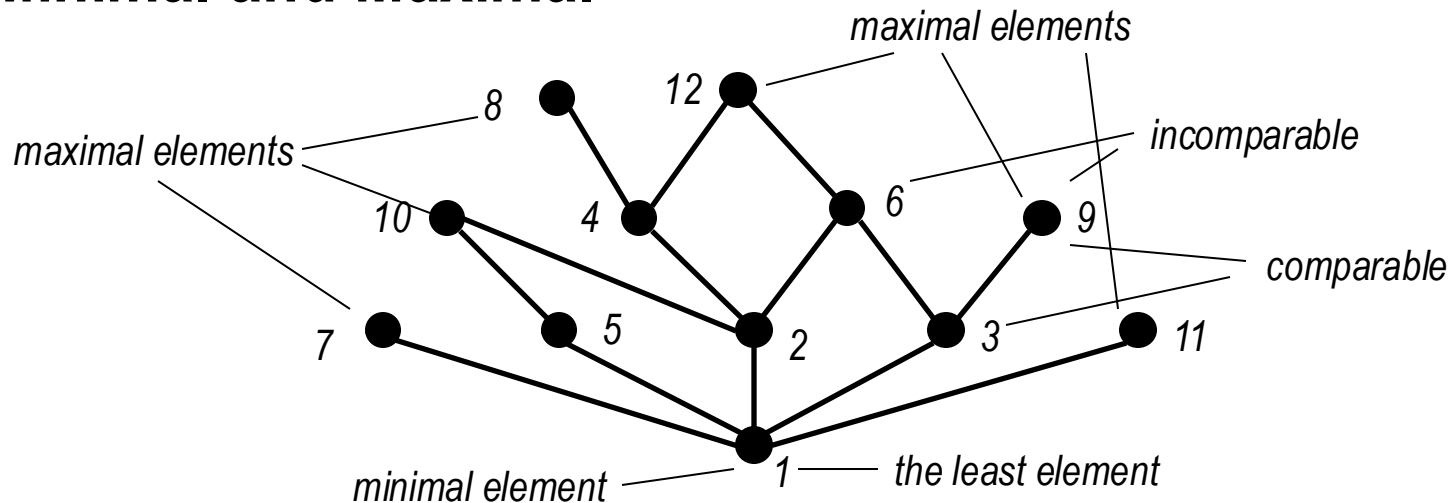
Anti-symmetric?

Divisibility, visualized

● Draw the divisibility order on $\{1, 2, \dots, 12\}$



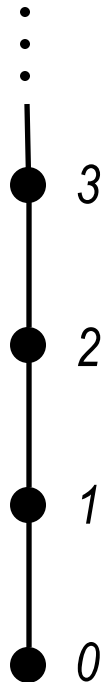
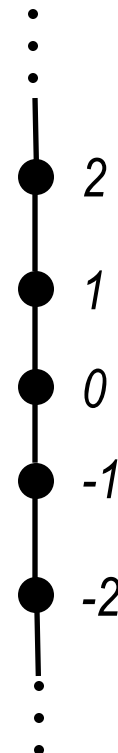
Minimal and Maximal



- Elements a, b are said to be **comparable** if $(a, b) \in R$ or $(b, a) \in R$
- Otherwise they are called **incomparable**
- Element a is **minimal** if for any b if $(b, a) \in R$ then $a = b$
- Element a is **maximal** if for any b if $(a, b) \in R$ then $a = b$
- Element a is called the **least element** if for any b , $(a, b) \in R$
- Element a is called the **greatest element** if for any b , $(b, a) \in R$

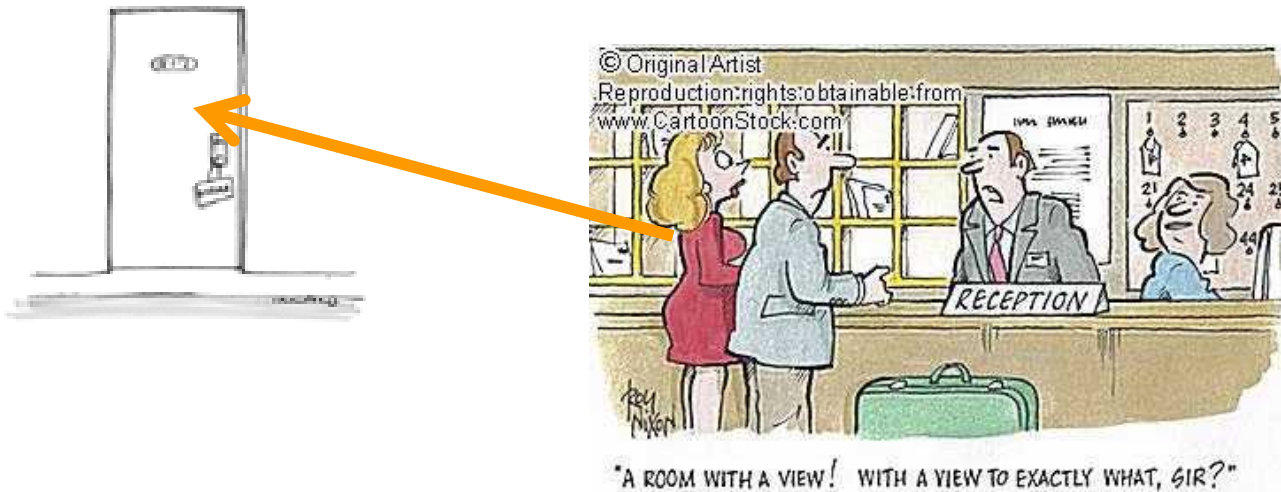
Total Order

- A partial order is **total** if every two elements are comparable
- Sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ are totally ordered with respect to $\leq$
- The power set of a set is **not** a total order with respect to $\subseteq$
- The diagram of a total order is a **chain**


 $\mathbb{N}$

 $\mathbb{Z}$


Functions

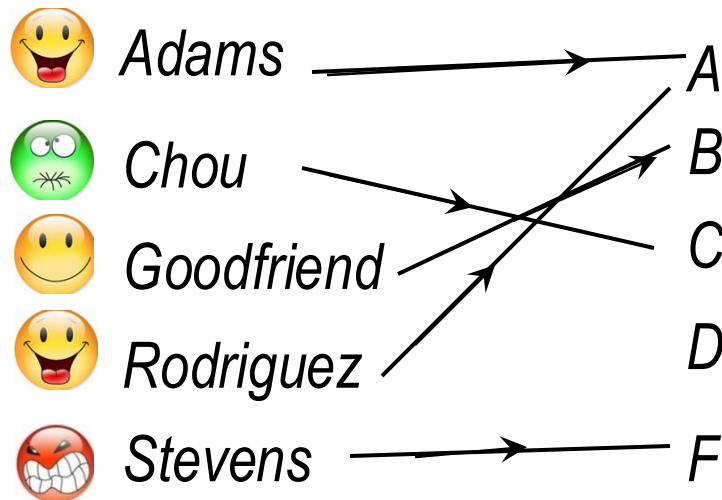
- Another important type of relation is a **function**
- Functions assign elements of one set to elements of another
- For example, assign rooms to people in a hotel



- Or we may assign a grade to each student from a class
- What we get is a set of pairs (Person, Door) or (Student, Grade), i.e. a relation, but a very particular one

Functions (cntd)

- A relation R from A to B is called a **function** from A to B , if for every $a \in A$ there is **exactly one** $b \in B$ such that $(a,b) \in R$.
 - Also called **mappings** or **transformations**



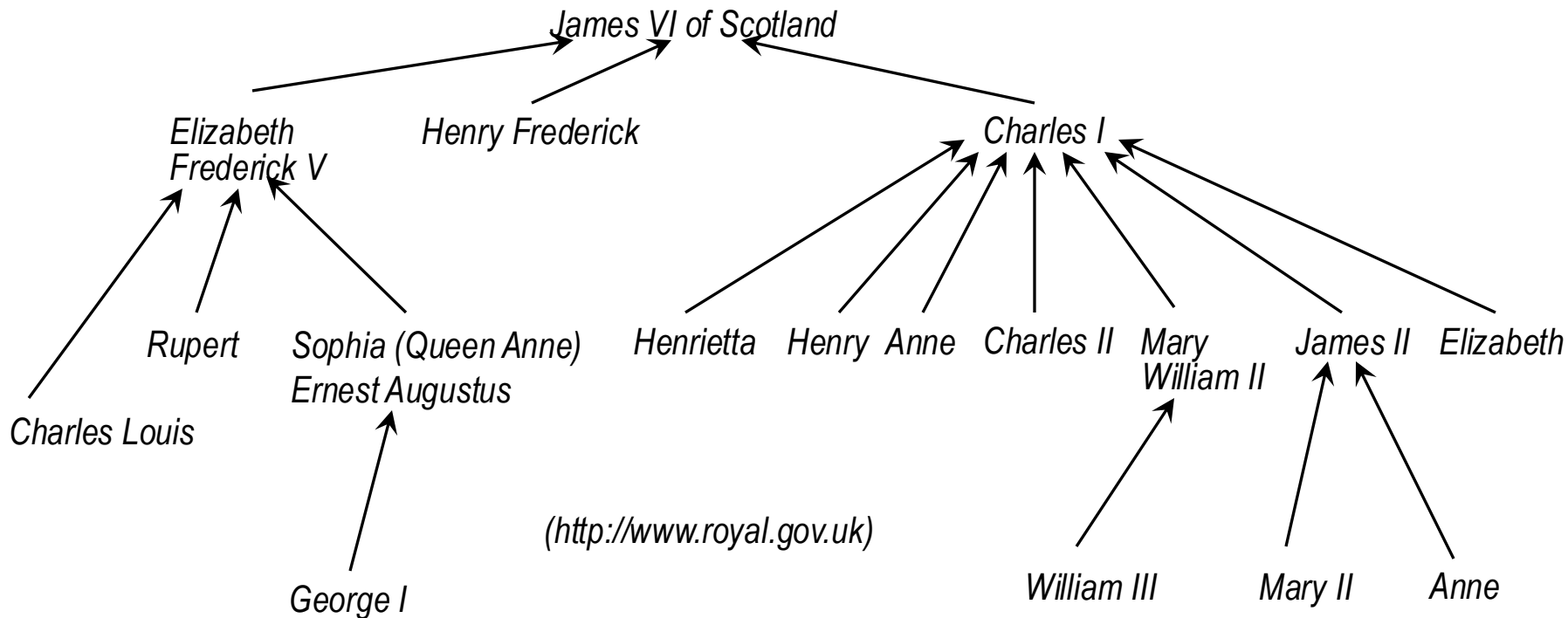
We use f, g, h to denote functions

$$f: A \rightarrow B \quad f(a) = b$$

$$f(\text{Rodriguez}) = A$$

Example

- Consider the relation $R = \{(a,b) \mid b \text{ is the (biological) father of } a\}$



- Is R a function?
- What about the relation $\{(a,b) \mid a \text{ and } b \text{ are brothers}\}$?

Describing Functions

- *A function is a relation, therefore we can use all methods of describing relations.*
- *Functions don't have many tuples though, so often use a table to list each pair of the relation instead.*

<i>Student</i>	<i>Grade</i>
<i>Adams</i>	<i>A</i>
<i>Chou</i>	<i>C</i>
<i>Goodfriend</i>	<i>B</i>
<i>Rodriguez</i>	<i>A</i>
<i>Stevens</i>	<i>F</i>

v.s.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>F</i>
<i>Adams</i>	1				
<i>Chou</i>			1		
<i>Goodfriend</i>		1			
<i>Rodriguez</i>	1				
<i>Stevens</i>					1

One 1 per row in matrix

Describing Functions (cntd)

- *Numerical functions can be computed using a formula*

$$f(x) = x^2$$

- *The most general way is to use some algorithm to compute a function*

`The letter grade is A, if the numerical mark is in between 100 and 85; the letter grade is B, if ...

Functions in programming languages:

*int **floor**(float real) {...}*

in Java

***function** floor(x: real): integer*

in Pascal

Domain and Codomain

- Let $f: A \rightarrow B$ be a function from A to B . Then A is called the **domain** of f , and B is called the **codomain** of f .



- If $f(a) = b$, then b is called an **image** of a , and a is called a **preimage** or **fiber** of b .
- How many images can a have via f ?
- How many preimages can b have?

Domain and Codomain (cntd)

● The **range** of f is the set of all images of elements of A

$$\text{range}(f) = \{ b \in B \mid \exists a \in A \ f(a) = b \}$$

● In our example:

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>F</i>
<i>Adams</i>	1				
<i>Chou</i>			1		
<i>Goodfriend</i>		1			
<i>Rodriguez</i>	1				
<i>Stevens</i>					1

$\text{domain} = \{ \textit{Adams}, \textit{Chou}, \textit{Goodfriend}, \textit{Rodriguez}, \textit{Stevens} \}$

$\text{codomain} = \{ A, B, C, D, F \}$

$\text{range} = \{ A, B, C, F \}$

Practice

- Are the following relations reflexive? symmetric? transitive? anti-symmetric?
 - Motherhood: ' x is the mother of y '
 - Intersect: ' $\text{straight lines } x \text{ and } y \text{ intersect}$ '
- Show that equivalence of propositional formulas is an equivalence relation on the set of propositional formulas
- Define a relation R on propositions such that (p,q) in R if and only if $p \rightarrow q$. Is this relation an equivalence? Is it an order? What if R is defined on **equivalence classes** of propositional formulas --- i.e. R is a relation on the set $S = \{ C(p) \mid p \text{ is a propositional formula} \}$ where $C(p)$ is the set of propositional formulas equivalent to p , and $(C(p),C(q))$ in R if $p \rightarrow q$?